AirWatch CO: Predictive Modelling of Urban Carbon Monoxide

MCA I Statistics Internal Assessment - Group 3 (Air Quality and Pollution)
Lead contributor: Farhan Sayed and Group 3
Dataset: UCI Air Quality (CNR-ISAC, Italy)
Target: next-hour CO(GT) = CO_next

1. Objective

Investigate how air pollution changes over time and how it relates to weather and sensor measurements. Build a model that predicts the next hour's carbon monoxide concentration CO(GT) using only information available up to the current hour, then explain usefulness and limitations for air-quality monitoring.

2. Dataset and Preprocessing

Cleaned records: 9353
Period: 2004-03-10 21:00:00 to 2005-04-04 13:00:00
Critical cleaning: hardware invalid codes (-200) treated as missing; NMHC(GT) dropped; time-aware imputation; chronological 70/15/15 split.
Train: 6547 | Val: 1403 | Test: 1403

3. Modelling Protocol

Features use only information available at hour t (sensors/weather, lags, calendar encodings). Validation uses TimeSeriesSplit (no random shuffle). Champion: Tuned HistGradientBoostingRegressor.
Best hyperparameters: {"l2_regularization": 0.5, "learning_rate": 0.1, "max_depth": 10, "max_iter": 100, "min_samples_leaf": 20, "random_state": 42}

4. Holdout Test Results

Model	Test RMSE	Test MAE	Test R2
Persistence Baseline	0.7739	0.4996	0.6556
Linear Regression	0.7556	0.6035	0.6717
Ridge Regression	0.7641	0.6156	0.6642
Random Forest	0.5842	0.4032	0.8037
Tuned HistGradientBoosting	0.5378	0.3608	0.8337

Champion: Tuned HistGradientBoostingRegressor | Test RMSE=0.5378 | MAE=0.3608 | R2=0.8337 | RMSE improvement vs baseline=30.5%

5. Usefulness and Limitations

Usefulness: short-horizon monitoring support, traffic-management hints, sensor sanity checks.
Limitations: single Italian station; historical 2004-2005 period; under-prediction on extreme high-CO hours; no explicit traffic counts; next-hour horizon only. Dashboard pollution tiers are project analytical bins (<2 / 2-4 / >4 mg/m3), not certified AQI.

6. Dashboard Enhancement

AirWatch CO presentation layer: FastAPI + React (Home, Explore, Insights, Predict, Evaluation, Team). Core academic work remains reproducible in Python notebooks. Local demo: start_project.ps1 / start_project.bat. Cloud configs: render.yaml + frontend/vercel.json (deploy when hosting credentials are available).

7. Member Contributions

Farhan Sayed (Data and Architecture Lead): preprocessing, API, dashboard, report packaging.
Member B (EDA and Analytics Lead): EDA statistics, diurnal/weekday analysis, correlation visuals. Replace with legal name before final submission.
Member C (Predictive Modelling Lead): CO_next target, baselines, TimeSeriesSplit tuning, artifacts. Replace with legal name before final submission.

8. Submission Bundle

Include: notebooks/01-04, data/cleaned/cleaned_air_quality.csv, artifacts/, reports/AIRWATCH_CO_FINAL_STATISTICS_REPORT.md|.html|.pdf, reports/VIVA_PRESENTATION_SCRIPT_AND_QA.md, README.md, backend/, frontend/.